

# Farshad Rahimi Ghashghaei

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🌐 [farshadrahimighashghaei.github.io](https://farshadrahimighashghaei.github.io) 📍 Khuzestan, Iran

## Profile

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- Highly motivated researcher in quantum information science with multiple publications in peer-reviewed scientific journals and presentations at international conferences.
- Cross-functional collaborator across the international institutions, resulting in successful interdisciplinary projects funded by Glasgow Caledonian University and Qatar National Library.
- Active research contributor and peer reviewer, resulting in demonstrated ability to produce, evaluate, and disseminate high-quality academic research within international scholarly communities.

## Education

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**M.Sc. in Cyber Security, Birmingham City University**

*Jan 2023 – Jan 2024  
Birmingham, United Kingdom*

- **Thesis title:** Enhancing the Security of Classical Communication with Post-Quantum Authenticated-Encryption Schemes for the Quantum Key Distribution
- **Supervisor:** Dr. Mehdi Yousefi
- **Grade:** Distinction (GPA: 4.0/4.0)

**B.Eng. in Computer Engineering, Institute for Higher Education ACECR Khuzestan**

*Sep 2018 – Sep 2022  
Khuzestan, Iran*

- **Supervisor:** Dr. Jamshid Afshani
- **Grade:** 13.82/20 (final research project: 19/20)

## Work Experience

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**System Administrator, Arya Ravesh Mahshahr Company**

*Nov 2020 – Dec 2022  
Khuzestan, Iran*

- Maintained and secured servers, including configurations and user access.
- Monitored performance, troubleshoot issues, and provided technical support.
- Documented IT processes, system configurations, and operating procedures for internal reference.
- Trained staff on best practices for cybersecurity and efficient use of resources.

## Teaching Experience

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**Workshop Instructor, Dana Mandegar Pars Institute**

*Feb 2025 – Jul 2025  
Lorestan, Iran*

- Delivered a 24-hour workshop on improving student focus and motivation.
- Conducted a 24-hour training on time management and classroom efficiency.
- Created customized teaching materials and interactive activities.
- Assessed participant progress and used feedback to refine workshops.

## Research & Publications

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### Journal papers:

- F. Rahimi Ghashghaei, N. Elmrabit, A.-U.-H. Qureshi, A. Akhunzada, and M. Yousefi, “Advanced Quantum Control With Ensemble Reinforcement Learning: A Case Study on the XY Spin Chain,” IEEE Access, vol. 13, pp. 49514–49526, Jan. 2025. doi: [10.1109/ACCESS.2025.3551232](https://doi.org/10.1109/ACCESS.2025.3551232).
- F. Rahimi Ghashghaei, Y. Ahmed, N. Elmrabit, and M. Yousefi, “Enhancing the Security of Classical Communication with Post-Quantum Authenticated-Encryption Schemes for the Quantum Key Distribution,” Computers, vol. 13, no. 7, pp. 163, Jul. 2024. doi: [10.3390/computers13070163](https://doi.org/10.3390/computers13070163).
- H. J. Jin, F. Rahimi Ghashghaei, N. Elmrabit, Y. Ahmed, and M. Yousefi, “Enhancing Sniffing Detection in IoT Home Wi-Fi Networks: An Ensemble Learning Approach with Network Monitoring System (NMS),” IEEE Access, vol. 12, pp. 86840–86853, Jun. 2024. doi: [10.1109/ACCESS.2024.3416095](https://doi.org/10.1109/ACCESS.2024.3416095).

### Conference papers:

- F. Rahimi Ghashghaei, “Recent Advances in Quantum Error Correction and Their Impact on the Stability of Quantum Computing,” 9th International Conference on Interdisciplinary Studies in Nanotechnology, Iran, 2026.
- F. Rahimi Ghashghaei, “Prediction of the Behavior of Many-Particle Systems in Quantum Mechanics Using Quantum Neural Networks,” 13th International Conference on Applied Research in Basic Sciences, Engineering and Technology, Georgia, 2026.

## Awards & Honors

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- Article selected as Editor’s Choice, Computers, 2024.
- Recognized as top students who achieved Distinction during the MSc course.

## Volunteer & Professional Activities

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### Peer Reviewer for following journals:

*Dec 2024 – Present*

- Quantum Reports: 1 review - [\[Certificate\]](#)
- Energies: 1 review - [\[Certificate\]](#)
- Journal of Computer Sciences and Informatics: 1 review - certificate available upon request.
- Symmetry: 2 reviews - [\[Certificate 1\]](#) - [\[Certificate 2\]](#)

## Technical Skills

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⚙️ **Quantum Simulation and Algorithms:** Experienced with simulating quantum systems, Familiar with variational algorithms (VQE, QAOA), and techniques for testing quantum operations.

🔒 **Quantum Cryptography:** Quantum key distribution (e.g., BB84), foundational knowledge of classical cryptography (hash functions, public-key schemes) and post-quantum cryptography (e.g., lattice-based schemes).

📊 **Machine Learning and Optimization:** Skilled in supervised, unsupervised, and reinforcement learning, with experience in gradient-based optimization and probabilistic modeling.

✂️ **Computational Methods:** Proficient in linear algebra, tensor network techniques (MPS/MPO), and stochastic simulation methods for modeling small systems.

🔧 **Programming and Tooling:** Python (NumPy, SciPy, Qiskit, QuTiP, PyTorch), C++, Linux, Git, L<sup>A</sup>T<sub>E</sub>X.

## Languages

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**English:** Advanced (C2)  
**German:** Intermediate (B1)  
**Spanish:** Beginner (A1)  
**Persian:** Native

## Referees

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Available upon request.